Physics-Guided Calibrated Localization of Methane Sources

Complete Technical Report

Project record — branch `final-benchmark`, data `csr-data-se48`

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1 Project overview

Goal. Convert methane super-emitter alerts into small, statistically guaranteed search regions from sparse networks of inexpensive sensors, under the wind-measurement error every real site has. The operational target is the U.S. EPA Super-Emitter Program's 50 m facility-association scale.

One-sentence result. On an exactly solvable testbed, direct physics inversion under a 10°/10% anemometer error returns the entire site (282 m radius, 0% at facility scale) at 90% guaranteed coverage, while three physics images fed to any of three set networks yield 61–75 m regions with up to 30% of scenarios inside 50 m — at the same per-scenario compute (≈420 ms on one CPU core) and with every paired improvement's 95% bootstrap interval excluding zero.

Method in brief. Three closed-form physics images per scenario (evidence, wind spread, visibility; K=8 Monte Carlo wind draws), a 10,433-parameter zero-initialized convolutional head joining any permutation-invariant set network at the logits, head frozen for the first half of training, ordinary cross-entropy, split conformal prediction at 90%.

2 The CH4-T simulator

500×500 m flat site; 64×64 candidate grid (cell ≈ 7.8 m); sources snapped to cell centers, positions uniform on $[0.1, 0.9]^2$.

Transport is the steady-state Gaussian plume with ground reflection:

$$g = \frac{1}{3600}\,\frac{1}{2\pi U \sigma_y(x)\sigma_z(x)}\, e^{-y_{\mathrm{cw}}^2/2\sigma_y^2}\left(e^{-(z_r-H)^2/2\sigma_z^2} + e^{-(z_r+H)^2/2\sigma_z^2}\right) \tag{1}$$

converted to ppm (1 ppm $CH_4 = 6.55\times10^{-7}\,\mathrm{kg\,m^{-3}}$), zero upwind ($x \le 0$); H=3 m, z_r=2 m; Briggs open-country σ_y, σ_z per Pasquill–Gifford class A–F (verified term-for-term against Hanna, Briggs & Hosker 1982); source-size floors 2 m/1 m; speed floor $0.5\,\mathrm{m\,s^{-1}}$. Mass conservation verified to 4×10^{-6}; crosswind symmetry to 4×10^{-14}.

Scenario draws (per scenario, seeded stream [root, split, i], root 20260717): rate $q \sim \mathrm{LogU}(100, 500)$ $\mathrm{kg\,h^{-1}}$ (super-emitters only); masts $S \sim \mathrm{U}\{4..8\}$ uniform on the site; stability uniform A–F; noise $\sigma \sim \mathrm{LogU}(0.2, 3.0)$ ppm; dropout $p_d \sim \mathrm{U}(0, 0.2)$ per reading. Wind: mean speed $\mathrm{LogU}(1, 8)\,\mathrm{m\,s^{-1}}$, uniform direction, AR(1) meander (ρ=0.7; direction amplitude $\mathrm{U}(20°, 40°)$; log-speed amplitude $\mathrm{U}(0.10, 0.30)$); T=30 instantaneous steady-plume snapshots (quasi-steady; no travel lag).

Observation. $y = q\,g_{c^*}(w) + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$ iid per (mast, step); readings are background-subtracted enhancement; sums downstream run over retained readings. Not modeled: bias, drift, correlated noise, terrain, multiple sources.

The information gap. The dataset records only the measured wind: per step, direction $+\mathcal{N}(0, (10°)^2)$ and speed $\times e^{\mathcal{N}(0, 0.10^2)}$ (empirical: 9.98°, 0.1000, no cross-step correlation). No model or map observes the true wind (verified bitwise: maps recomputed at the measured wind match stored artifacts to 0.0; at the true wind they differ with total variation 1.0).

Splits. 10,000 train / 1,000 val / 2,000 calib / 2,000 test; content-hash disjoint on all six pairs; regeneration reproduces stored files bit-for-bit. Why time matters: a steady-wind predecessor testbed left 57% of scenarios weakly identifiable; the meander reduces this to ∼8% (2–8-mast probe) — the sweep supplies bearing/time-of-arrival information.

3 Method

3.1 Physics images (the "ens" stack)

For each cell c and draw $w^{(k)} \sim \mathcal{E}_{10°,10\%}(\hat{w})$, k=1..8 (stream 92): $a_c = g_c^\top y$, $b_c = \|g_c\|^2$, $z_c = a_c/(\sigma\sqrt{b_c})$, computed *forward* from every candidate cell (no inversion anywhere; the only per-cell solve is the scalar best rate

$\hat{q}_c = \max(0, a_c/b_c)$). Channels:

$$\text{evidence: } \frac{1}{K}\sum_k \operatorname{asinh}(z_c^{(k)}/10) \qquad \text{wind spread: } 3\,\mathrm{std}_k \operatorname{asinh}(z_c^{(k)}/10)$$
$$\text{visibility: } \frac{1}{K}\sum_k (\log_{10} b_c^{(k)} + 8)/6$$

The asinh replaces a hard clip that saturated ≈40% of near-source neighborhoods on this population (train-split measurement); it equals $z/10$ for small signals and never plateaus. The spread channel deliberately uses the Monte Carlo sampling variability as a feature (wind fragility), not as error to suppress; the analytic direction-smeared alternative (zero MC noise) was tested and performed worse.

3.2 Networks

Shared token encoding (Fourier of mast x, y and t; $\operatorname{asinh}(y/\sigma)$; instantaneous wind), shared context MLP ($u, v, \log D, \log\sigma, S$), shared decoder (→1024→4096 logits); only aggregation differs: DeepSets (masked mean+max pool; 4,870,080 params), a custom message-passing graph network (mast nodes; ordered edges carrying pair geometry and the mean and max wind alignment $u\cdot\hat{\Delta}$; two rounds; 5,610,048), and a Set Transformer (2×ISAB, 4 heads, 16 inducing points, PMA(1); 6,528,448). Head (identical for all): Conv 3→32→32→1 (3×3, GELU), final layer zero-initialized, 10,433 params (+0.16–0.21%), joined at the logits $\ell = f_\theta(x) + h_\phi(M(x))$.

3.3 Training

AdamW, lr $3\times10^{-4}\to3\times10^{-5}$ cosine, wd 10^{-4}, batch 256, 200 epochs, clip 1.0, bfloat16, D4 augmentation (jointly on sensors, source, wind, and map channels), targets = 0.75-cell-blurred one-hot, best validation-NLL checkpoint, seeds 1–3, no per-model tuning. **Head warmup:** head parameters frozen for epochs 0–99, joining from zero at 100. Rationale (measured): joint training lets the head's easy early signal displace base feature learning — jointly trained GNN/ST reached *worse* validation loss with maps than without and worse regions than their own baselines; a prespecified pilot on a disjoint seed recovered the GNN from 72–74 m to 61.7 m. Verified property: warmed runs reproduce their baseline's validation trajectory to every printed digit through epoch 75 (12/15 runs; the three ST exceptions trace to CUDA attention nondeterminism, orthogonal to the comparison).

3.4 Conformal calibration (after training, never during)

Score = randomized cumulative probability mass needed to include the true cell; threshold = $\lceil (n+1)(1-\alpha) \rceil$-th smallest of n=2000 calibration scores (α=0.1 ⇒ k=1801); region = smallest top-probability set reaching the threshold. Guarantee rests on calibration/test exchangeability; training, checkpoint selection, and calibration use disjoint splits (verified in code and by artifact mtimes). Known nuance: the randomized-score/deterministic-region pair over-covers only near-atomic maps (affects the exact reference: empirical 0.958); immaterial for the diffuse learned maps (coverages 0.887–0.916).

4 Results (final benchmark: 4–8 masts, $q \geq 100$, measured wind)

4.1 Main table (3 seeds; min–max)

Method	Coverage	Radius (m)	<50 m
Wind-averaged physics only (K=8)	1.000	282 (full site)	0%
DeepSets	0.887–0.914	97–100	0–0.1%
DeepSets + maps	0.890–0.916	**73–75**	**16–17%**
GNN	0.896–0.904	68–69	1–7%
GNN + maps	0.889–0.911	**61–62**	**27–29%**
Set Transformer	0.891–0.901	72–75	8–12%
Set Transformer + maps	0.887–0.906	**64–65**	**28–30%**

Exact-model reference under *known* wind: 4.4 m (cov 0.958, 75.5% <50 m) — the price of wind error and sparse sensing is the distance from 4.4 to ∼62–75.

4.2 Paired effects (seed 1, 10^4 paired bootstrap)

DeepSets -24.6 m $[-25.8, -23.0]$, $+16.2$ pp $[+14.6, +17.9]$, smaller in 95.9% of scenarios; GNN -5.2 $[-6.4, -4.2]$, $+20.5$ $[+18.6, +22.3]$, 83.2%; ST -10.7 $[-12.1, -9.5]$, $+20.9$ $[+19.1, +22.7]$, 87.4%. All six intervals exclude zero.

4.3 Component ladder (DeepSets, 3 seeds, monotone per seed)

Input-only 97–100 m / 0% → +evidence 75–78 / 11–12% → +visibility 74–76 / 12–14% → +wind spread 73–75 / 16–17%. Evidence is the dominant rung (-21 to -23 m per seed). A random-map negative control matches the no-maps baseline, so the gains are physics content, not capacity.

4.4 Sensor density and operating points (seed 1)

Median / <50 m by mast count: with maps, GNN goes 87 m/14% (4 masts) → 55 m/38% (8 masts); all six configurations improve monotonically with density, and with-maps beats no-maps at every count for every architecture. At the 80% confidence level (labeled triage point): GNN and ST reach ≈52 m with ≈46% <50 m; the 90% level is the headline throughout.

4.5 Inference cost (Intel Xeon Gold 6426Y, 1 thread, batch 1, median)

Physics-only inversion 420 ms; original networks 2/3/7 ms (DS/GNN/ST); with maps 414/425/418 ms, of which the three images are ≈410–420 ms and the forward pass 4/5/9 ms. Same physics budget as the failed inversion; ≈0.02% of a 30-minute window.

4.6 Explainability (exact Shapley values over the three maps)

Value function: log-probability of the true cell; 2^3 coalitions enumerated exactly; seed-1 checkpoints; 2,000 scenarios per model. Mean ϕ (nats): evidence $+0.79/+0.37/+0.60$ (DS/GNN/ST), visibility $+0.10/+0.01/+0.05$, wind spread $-0.11/-0.05/-0.10$. The spread map's slightly negative mean in nats coexists with its positive effect on regions (the ladder): it trades confidence at the truth for suppression of wind-fragile false peaks — negative in nats, positive in metres.

5 Findings beyond the headline

Sensor-density boundary. On a denser 6–12-mast population (complete 18-run grid, archived), the same maps *hurt* the relational architectures' medians (GNN 54.5→62.6 m) while all with-maps models converged to a common band — physics inputs substitute for sensor density and stop paying when raw data suffices.

Evidence-map saturation. The original clipped evidence channel saturated ≈40% of near-source neighborhoods on the super-emitter population, imposing an 81–83 m ceiling on all with-maps models; replacing the clip with asinh dropped the ceiling ≈8 m and raised facility-scale rates ≈5–8 pp (archived before/after).

Feature displacement and the warmup. See §3.3 — hypothesized from validation curves, confirmed by a prespecified pilot, replicated across seeds and architectures; an instance of the staged-training / feature-distortion principle (gradual unfreezing; linear-probe-then-fine-tune).

Negative results retained. Seed ensembling: no gain (correlated errors). Analytic smeared maps: worse than the MC ensemble. Physics-residual training losses (earlier phase): useless to mildly harmful. Exact wind-marginalized oracle: intractable (documented quadrature study).

6 Verification record

Three independent adversarial audits on the final state (plus earlier rounds): (i) data/leakage — split disjointness 0/6 pairs, population conformance, bitwise map and wind reproducibility, oracle prior verified numerically on $[100, 500]$; (ii) training fairness — warmup unit checks (AdamW leaves frozen zero head untouched), first-100-epoch trajectory equivalence, artifact/tag hygiene, identical settings; (iii) results — every package number recomputed from per-scenario audit arrays (48,000 mask-integrity row-checks, 0 mismatches), independent re-bootstrap of all CIs, figure claims re-verified at 5 quantiles and every mast count. The conformal-on-exact self-calibration gate passes ($0.958 \in [0.85, 0.985]$, bound calibrated by a pre-launch review measurement).

7 Reproducibility and provenance

Artifacts (`csr-data-se48`): four splits, measured winds, det/ens maps, calib/test oracles, 24 audit npzs (sizes + covered + packed masks), all 24 training checkpoints. **Results**: `results/paired_wind.json` and `se48_table_deltas_ci.json` with its generating script, plus `results_package/` (tables PDF, technical reference, figures). **Scripts**: datagen/chains/ladder (idempotent, atomic writes, CUDA preflight, env-pinned), table/CI/figure generators, the SHAP pipeline. **Decision log**: `GATES.md` (dated entries for every gate and every adaptive decision) plus `SUPER_EMITTER_CHANGES.md`. **Archived superseded configurations** (all retained with results): full population; 6–12 grid; 2–8 partial; clipped-evidence era; joint-training era incl. the warmup pilot.

8 Limitations and required disclosures

(1) Simulation-only; the maps use the generating forward model; field validation is future work. (2) Idealizations: quasi-steady plume, flat terrain, single steady source, known stability class and noise scale, iid noise. (3) The benchmark population (4–8 masts, $q \geq 100$) and two method refinements (asinh; warmup) were selected adaptively during development — disclosed, pilot-validated where applicable, with every before-state archived; effect *directions* are more robust than exact magnitudes. (4) The wind-error budget is taken as the instrument specification; representativeness error is subsumed into it; sensitivity to a misspecified budget is untested. (5) Set Transformer training is not bit-reproducible (CUDA attention); reported as 3-seed ranges. (6) The 90% guarantee is marginal over the scenario population, not per-site conditional; subgroup coverage by mast count is the corresponding diagnostic. (7) Roughly 70% of alerts still do not resolve to 50 m at 90% confidence; the density curve (4→8 masts) is the deployment lever.

9 Figure and asset inventory

`figures/paper`: `fig_data_dist` (region-radius ECDFs), `fig_data_masts` (median vs mast count), `fig_mc_simple` (3-step Monte Carlo process), `fig_simulator` (cumulative exposure spreading, 3 boxed panels), `fig_montecarlo`, `fig_spread_signal` (spares); the four map visuals; pipeline diagrams. `Poster Graphics v3`: XAI set (attributions, UMAP/PCA, integrated gradients) and `shap/shap_beeswarm_models` (per-model beeswarm, ±3 nats, feature-value coloring; pptx export). All figure scripts regenerate their outputs from the live artifacts.